

Chemistry — Assignment 7: Isomerism & Threo/Erythro

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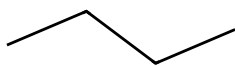
Chemistry (Foundational Chemistry) · Semester 5

Question. What is isomerism? What is threo and erythro? Classify isomerism with examples.

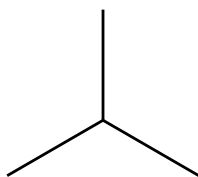
Isomerism: compounds with the same molecular formula but a different arrangement of atoms (structure) or spatial orientation.

Structural (constitutional): chain, position, functional, metamerism, tautomerism. **Stereoisomerism:** geometrical (cis/trans, E/Z) and optical (R/S; enantiomers & diastereomers).

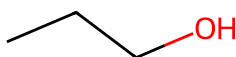
Threo / erythro describe the relative configuration of two adjacent stereocentres: **erythro** = like groups on the *same* side in the Fischer projection (meso-like); **threo** = like groups on *opposite* sides (the $\pm$) pair).



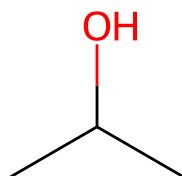
n-butane



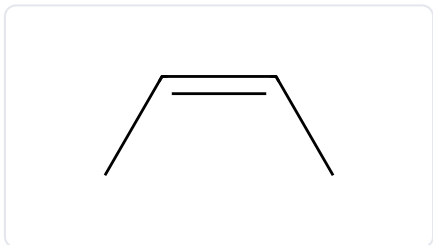
isobutane (chain isomer)



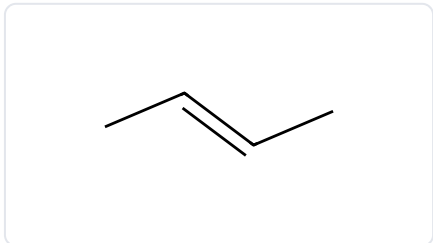
propan-1-ol



propan-2-ol (position isomer)



cis-but-2-ene



trans-but-2-ene